ÉCOLE UNIVERSITAIRE DE PHYSIQUE ET D'INGÉNIERIE Université Clermont Auvergne	**2022-2023** M2PFA UP & NP IMMAP DU Data Scientist
Statistics Projects	

Central Limit Theorem at fail: the Cauchy distribution.

Let X and Y be two independent and identically distributed Normal Random Variables :

$$\mathcal{N}_{\mathrm{X}}(x; \bar{\mu}_{\mathrm{X}} = 0, \bar{\sigma}_{\mathrm{X}} = 1) \sim \mathcal{N}_{\mathrm{Y}}(y; \mu_{\mathrm{Y}} = 0, \bar{\sigma}_{\mathrm{Y}} = 1).$$

We'll consider successively the random variables Z built as the sum (Z=X+Y), the product (Z=XY) and the ratio (Z=X/Y) of these two variables.

1 Sum of two Normal Variables

Let's consider Z, the sum of two Normal Variables: $\mathrm{Z} = \mathrm{X} + \mathrm{Y}$.

1. Derive the expected value, $\bar{\mu}_{\mathrm{Z}} = \mathbb{E}[\mathrm{X} + \mathrm{Y}]$, and the variance, $\bar{\sigma}^2_{\mathrm{Z}} = \mathbb{V}[\mathrm{X} + \mathrm{Y}]$.

2. Generate a sample of $\mathrm{N} = 10^4$ observations of Z and build the corresponding histogram, with 100 bins in the range $\pm 5\sqrt{S^2_{\mathrm{N}}}$, where S^2_{N} is the sample variance.

3. Using the Least-Squares method [1], perform a binned fit to the histogram with the Gaussian profile:

   $$\mathcal{P}_{\mathrm{Z}}(z; \mu_{\mathrm{Z}}, \sigma_{\mathrm{Z}}) = \mathcal{N}_0 \cdot \mathcal{G}_{\mathrm{Z}}(z; \mu_{\mathrm{Z}}, \sigma_{\mathrm{Z}}),$$

   where the parameter $\mathcal{N}_0$ is an overall normalisation factor and

   $$\mathcal{G}_{\mathrm{Z}}(z; \mu_{\mathrm{Z}}, \sigma_{\mathrm{Z}}) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{z-\mu_{\mathrm{Z}}}{\sigma_{\mathrm{Z}}}\right)^2}$$

   is the Gaussian probability density function (PDF). Give the best-fit parameters $\hat{\mathcal{N}}_0$, $\hat{\mu}_{\mathrm{Z}}$ and $\hat{\sigma}_{\mathrm{Z}}$ with their uncertainty. Are they consistent with their expectation [2] ?

4. We now consider the mean of n Random Variables Z :

   $$M_{\mathrm{Z}}(n) = \frac{1}{n}\sum_{k=1}^{n} z_k.$$

   Derive the expected value, $\bar{\mu}_{M_{\mathrm{Z}}} = \mathbb{E}[M_{\mathrm{Z}}]$, and the variance, $\bar{\sigma}^2_{M_{\mathrm{Z}}} = \mathbb{V}[M_{\mathrm{Z}}]$ as a function of n.

5. For different values of n varying from 1 to 101 (e.g. 25 steps of 4 units), generate the samples of $\mathrm{N} = 10^4$ realizations of $M_{\mathrm{Z}}(n)$ and extract the parameters $\hat{\mu}_{M_{\mathrm{Z}}}(n)$ and $\hat{\sigma}_{M_{\mathrm{Z}}}(n)$ from a gaussian fit to the corresponding histograms (100 bins in the range $\pm 5\sqrt{S^2_{\mathrm{N}}/n}$).

[1] Instructions given in appendix.

[2] The normalisation factor $\mathcal{N}_0$ multiplied by the bin width should be consistent with the sample size N modulo the (few) observations falling outside the histogram range.

(a) Produce the graph of $\hat{\mu}_{M_{\rm Z}}(n)$ (with error-bars) as a function of n. Does the estimator $\hat{\mu}_{M_{\rm Z}}$ correctly converge to the expected value $\bar{\mu}_{M_{\rm Z}}$ when n increase?

(b) Produce the graph of $\hat{\sigma}_{M_{\rm Z}}(n)$ (with error-bars) as a function of n. Perform a fit to the measured points with a power-law function $\hat{\sigma}_{M_{\rm Z}}(n;\sigma_0,\alpha)=\sigma_0 n^{\alpha}$. Are the best-fit parameters, $\hat{\sigma}_0$ and $\hat{\alpha}$ consistent with the expected behaviour: $\hat{\sigma}_{M_{\rm Z}}(n)=\frac{\hat{\sigma}_{\rm Z}}{\sqrt{n}}$?

2 Product of two Normal Variables

Let's now consider the product of two Normal Variables $\mathrm{Z}=\mathrm{XY}$. The theoretical probability density function of Z is given by the *Normal product distribution*:

$$\mathcal{B}_{\rm Z}(z)=\frac{1}{\pi\sigma_{\rm X}\sigma_{\rm Y}}\mathcal{K}_0\left(\frac{|z|}{\sigma_{\rm X}\sigma_{\rm Y}}\right),$$

where $\mathcal{K}_\nu(z)$ is the modified Bessel function of the second kind. The distribution presents a very peaked symmetrical profile with a (logarithmic) divergence in $z=0$.

1. Verify that the expected value vanish, $\bar{\mu}_{\rm Z}=\mathbb{E}[\mathrm{XY}]=0$, and the variance, $\bar{\sigma}^2_{\rm Z}=\mathbb{V}[\mathrm{XY}]=1$[3].

2. Generate a sample of $\mathrm{N}=10^4$ Random Variables $\mathrm{Z}=\mathrm{XY}$ and display the corresponding histogram (100 bins in the range $\pm5\sqrt{S^2_{\rm N}}$)

3. Using the Least Squares method, perform a binned fit to the histogram with a Bessel-like profile:

$$\mathcal{P}_Z(z;z_0,a)=\frac{\mathcal{N}_0}{\pi a}\cdot\mathcal{K}_0\left(\left|\frac{z-z_0}{a}\right|\right),$$

   where the parameter $\mathcal{N}_0$ is an overall normalisation factor and $\mathcal{K}_0(z)$ is the modified Bessel function discussed above [4]. Extract the best-fit parameters $\hat{\mathcal{N}}_0$, $\hat{z}_0$ and $\hat{a}$ with their uncertainty and verify that the location parameter $\hat{z}_0$ and the scale parameter $\hat{a}$ are consistent with their theoretical value: $z_0=0$ and $a=\bar{\sigma}_{\rm X}\bar{\sigma}_{\rm Y}=\bar{\sigma}^2_{\rm Z}=1$.

4. We consider the mean of n Random Variables Z. For different value of n varying from 1 to 101 (25 steps of 4 units), generate the samples of $N=10^4$ Random Variables $M_{\rm Z}(n)$ and produce the corresponding histograms (100 bins in the range $\pm5\sqrt{S^2_{\rm N}/n}$). Check that for large enough n values ($n\gtrsim25$), the Variable $M_{\rm Z}(n)$ tends to exhibit a gaussian distribution, as a consequence of the Central Limit Theorem.

5. For $n\geq25$, extract the parameters $\hat{\mu}_{M_{\rm Z}}(n)$ and $\hat{\sigma}_{M_{\rm Z}}(n)$ from a gaussian fit to the histograms.

   (a) Produce the graph of $\hat{\mu}_{M_{\rm Z}}(n)$ (with error-bars) as a function of n.

   (b) Produce the graph of $\hat{\sigma}_{M_{\rm Z}}(n)$ (with error-bars) as a function of n. Perform a fit to the graph points with the power-law function $\hat{\sigma}_{M_{\rm Z}}(n)=\sigma_0 n^{\alpha}$ and check the obtained parameters, $\hat{\sigma}_0$ and $\hat{\alpha}$ are consistent with the expected behaviour.

[3] You can easily demonstrate: $\mathbb{V}[\mathrm{XY}]=\mathbb{E}[X^2Y^2]-\mathbb{E}[XY]^2=(\mathbb{V}[\mathrm{X}]+\mathbb{E}[\mathrm{X}]^2)(\mathbb{V}[\mathrm{Y}]+\mathbb{E}[\mathrm{Y}]^2)-(\mathbb{E}[\mathrm{X}]\mathbb{E}[\mathrm{Y}])^2$ when both (X,Y) and (X^2,Y^2) are pairs of uncorrelated variables.

[4] The scipy module provides this modified Bessel functions as *scipy.special.kn*$(0,z)$.

3 Ratio of two Normal Variables : the Cauchy law

Let's now consider the ratio of two Normal Variables Z = X/Y. The theoretical probability density function of Z is given by the *Cauchy law*:

$$\mathcal{C}_{\mathrm{Z}}(z; z_0, \Gamma) = \frac{1}{\pi\Gamma}\left[1 + \left(\frac{z - z_0}{\Gamma}\right)^2\right]^{-1},$$

that exhibits a symmetrical *Lorentzian* profile with very long tails. The location parameter z_0 is the *median* of the distribution, and the scale parameter Γ is the *width*. The Cauchy law has the specific property to have no moment generating function: the expected value of a Cauchy-distributed variable is undefined and all the higher-order moments diverge. This property prevents the Central Limit Theorem to apply to the Cauchy distribution, as illustrated with this exercise.

1. Justify that the expected value, $\bar{\mu}_{\mathrm{Z}} = \mathbb{E}[\mathrm{X}/\mathrm{Y}]$ is undefined, and the variance, $\bar{\sigma}^2_{\mathrm{Z}} = \mathbb{V}[\mathrm{X}/\mathrm{Y}]$ tends to infinity.

2. Generate a sample of N = 10^4 Random Variables Z = X/Y and display the corresponding histogram with 100 bins in the range ± 50 [5].

3. Using the Least Squares method, perform a binned fit to the histogram with the Cauchy distribution:
$$\mathcal{P}_Z(z; z_0, \Gamma) = \mathcal{N}_0 \cdot \mathcal{C}_{\mathrm{Z}}(z; z_0, \Gamma),$$
where the parameter $\mathcal{N}_0$ is an overall normalisation factor and $\mathcal{C}_{\mathrm{Z}}(z; z_0, \Gamma)$ the Cauchy PDF. Extract the best-fit parameters $\hat{\mathcal{N}}_0$, $\hat{z}_0$ and $\hat{\Gamma}$ with their uncertainty.

4. We consider the mean of n Random Variables Z. Check that for a large value of n (e.g. $n = 50$), the Variable $M_{\mathrm{Z}}(n)$ is still described by the same Cauchy function with parameters close to the ones for $n = 1$.

5. For different value of n varying from 1 to 101 (25 steps of 4 units), generate the samples of $N = 10^4$ Random Variables $M_{\mathrm{Z}}(n)$ and extract the parameters $\hat{z}_{0_{M_{\mathrm{Z}}}}(n)$ and $\hat{\Gamma}_{M_{\mathrm{Z}}}(n)$ from a Cauchy-law fit to the corresponding histograms (100 bins in the range ± 50).

   (a) Produce the graph of $\hat{z}_{0_{M_{\mathrm{Z}}}}(n)$ (with error-bars) as a function of n.

   (b) Produce the graph of $\hat{\Gamma}_{M_{\mathrm{Z}}}(n)$ (with error-bars) as a function of n. Perform a fit to the graph points with the power-law function $\hat{\Gamma}_{M_{\mathrm{Z}}}(n) = \Gamma_0 n^{\alpha}$.

   (c) Conclude.

4 Appendix

1. **Histogram fitting**
   The scipy method:
$$pars, cov = scipy.optimize.curve_fit(f, xdata, ydata, p0, sigma, method =' lm')$$
   provides an useful tool for a Least Squares fit to data points. It directly returns the vector of the fitted parameters (*pars*) and the corresponding covariance matrix (*cov*). When fitting a binned

[5] You can check by increasing N to very high value that the sample variance diverges for this distribution and that some observations fall far outside the histogram range. Use a log-scale to better vizualize the large tails of the Cauchy distribution compared to the Gaussian one

sample (i.e. an histogram) the data points coordinates are defined by the center of the bins ($xdata$) and the bins content ($ydata$). In order to have a reliable estimation of the covariance matrix of the fit parameters you must define the uncertainty associated to the content of each bin ($sigma$). The number of entries in each bin is usually considered as a Poisson variable and a confidence interval can be defined accordingly. Let n_k be the content of the kth histogram bin. The (asymmetric) uncertainty interval $[l_k, u_k]$ surrounding the observed value n_k with a Confidence Level $p = 1 - 2\alpha$ can be derived from the percent-point function of the χ^2-law ($\text{PPF}_{\chi^2}(\alpha, ndof)$), as :

$$\begin{aligned} l_k &= \text{PPF}_{\chi^2}(\alpha, 2n_k)/2 \\ u_k &= \text{PPF}_{\chi^2}(1-\alpha, 2(n_k+1))/2 \end{aligned}$$

that can easily be implemented using the method:

$$scipy.stats.chi2.ppf(\alpha, ndof).$$

The usage is to define the intervals for a given Confidence Level $p = 68.3\%$ (corresponding to $\alpha = 0.158$), to be consistent with the one-sigma standard deviations that usually defines the gaussian intervals. Note that for a n_k value large enough ($n_k \gtrsim 20$), the Poisson variable is almost gaussianly distributed, thanks to the Central Limit behaviour, and the $p = 68.3\%$ confidence interval can be approximated by the almost-symmetrical range, $n_k \pm \delta_{n_k}$, with $\delta_{n_k} \simeq \sigma_{n_k} = \sqrt{n_k}$. For smaller bin contents, the asymmetrical interval $[l_k, u_k]$ defined above must be used, with a special care for the empty bins ($n_k = 0$) for which only the upper limit u_k can be defined (the number of entries cannot be negative !). Unfortunately, the *curve_fit* method cannot deal with asymmetrical uncertainties. I suggest to define a symmetrized interval $n_k \pm \delta_{n_k}$ for all bins, where the uncertainty δ_{n_k} is either the *averaged* error, $\delta_{n_k} = (u_k - l_k)/2$, or the most conservative error: $\delta_{n_k} = max(u_k - n_k, n_k - l_k)$. For empty bins the uncertainty can be set to $\delta_{n_k=0} = u_{k=0} = \text{PPF}_{\chi^2}(1 - 0.158; 2)/2 = 1.84$.

2. **Alternative fitting**
   Alternatively you can decide to perform a more precise unbinned fit to the generated data sample by implementing a log-likelihood minimization. This can be done with the dedicated

$$scipy.stats.{<}rv_continuous{>}.logpdf()$$

   methods for instance. In that case, you will have to explore the log-likelihood function to derive the confidence interval for each of the parameters (e.g. with the $-\Delta\text{logL}(\mu_\pm) = 1/2$ method, neglecting any correlation between the parameters). You are also free to use another python package to perform the fit and evaluate the parameters uncertainties.

3. **Suggestions**
   The three parts of the problem are almost identical: only the Random Variable is different. Defining generic methods to generate the samples, producing the histograms, evaluating the uncertainties, and performing the fits would help ...
   Do not forget to add comments in the python script to describe and explain the different steps of the analysis.
   Once the script is complete you can repeat the exercices using larger samples (e.g. $\text{N} = 10^5$ or even larger) to get more precise results (slow and CPU-demanding processing ...)